



Cambridge IGCSE™

BIOLOGY

0610/62

Paper 6 Alternative to Practical

May/June 2023

MARK SCHEME

Maximum Mark: 40

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the May/June 2023 series for most Cambridge IGCSE, Cambridge International A and AS Level and Cambridge Pre-U components, and some Cambridge O Level components.

PUBLISHED**Generic Marking Principles**

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Science-Specific Marking Principles

1 Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.

2 The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.

3 Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).

4 The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.

5 'List rule' guidance

For questions that require ***n*** responses (e.g. State **two** reasons ...):

- The response should be read as continuous prose, even when numbered answer spaces are provided.
- Any response marked *ignore* in the mark scheme should not count towards ***n***.
- Incorrect responses should not be awarded credit but will still count towards ***n***.
- Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should **not** be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response.
- Non-contradictory responses after the first ***n*** responses may be ignored even if they include incorrect science.

6 Calculation specific guidance

Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'.

For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values.

For answers given in standard form (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme.

Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.

7 Guidance for chemical equations

Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme.

State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

Mark scheme abbreviations

- ; separates marking points
- / alternative responses for the same marking point
- **R** reject the response
- **A** accept the response
- **I** ignore the response
- **ecf** error carried forward
- AVP any valid point
- **ora** or reverse argument
- AW alternative wording
- underline actual word given must be used by candidate (grammatical variants excepted)
- () the word / phrase in brackets is not required but sets the context

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Question	Answer	Marks	Guidance
1(a)(i)	68.5 (°C) ;	1	A 68 / 69
1(a)(ii)	table drawn with at least four, columns / rows and a header line ; two suitable headings with units ; all three sets of data recorded (14 values) ;	3	MP2 temperature / °C and time / minutes OR temperature / °C and number of (test-)tubes
1(a)(iii)	axes labelled with units ; suitable linear scale and data occupies at least half the grid in both directions ; 14 points plotted to an accuracy of $\pm$ half a small square ; suitable line drawn for each of the three data sets ; key correctly identifies each of the three data sets ;	5	
1(a)(iv)	huddling, reduces heat loss / conserves more heat ; ora the more test-tubes the slower, the temperature drops / heat loss ora OR as time increases, temperature falls / heat is lost ;	2	
1(a)(v)	<i>any two from:</i> (same) size test-tubes ; (same) type / kind, of test-tubes ; all test-tubes placed in a beaker ; (same) total experimental time / described ; (same) time intervals (between sampling) ;	2	

Question	Answer	Marks	Guidance
1(a)(vi)	<i>any two from:</i> volume of water, not the same / could vary / AW ; temperature (of water) not measured at , start / 0 minutes ; start temperature, not the same / could vary / AW ; no band around 1 test-tube/ bands only on 3 & 7 test-tubes OR no insulating effect in 1 test-tube / band on 3 & 7 test-tubes absorbs heat ; band tightness, not the same / could vary / AW ; room / environmental / surrounding, temperature, not the same / could vary / AW ;	2	
1(a)(vii)	<i>any one from:</i> repeat the whole investigation (at least) twice more / AW ; same volume of water ; temperature (of water) measured at , start / 0 minutes ; same starting temperature ; band around 1 test-tube ; band tightness the same ; maintain the, room / environmental , temperature ;	1	
1(a)(viii)	(use of) hot, water / test-tubes / equipment OR (use / breakage, of) glass (equipment) ;	1	

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Question	Answer	Marks	Guidance
1(b)	<i>either – reducing sugar method</i> <i>independent variable</i> 1 at least three temperature values, tested / stated ; <i>dependent variable</i> 2 time to (first) (correct) colour change (of Benedict's) / (correct) colour (of Benedict's) after a set period of time / mass of precipitate in a fixed time ; 3, 4 and 5 variables kept constant ;;; concentration of , amylase / enzyme (solution) volume / mass, of, amylase / enzyme (solution) pH volume of starch (solution) / mass of starch / mass of plant material / size of plant material concentration of starch (solution) 6, 7 and 8 method ;;; equilibration (both solutions in water bath before mixing) take (fixed volume of) samples (testing with) Benedict's (solution) use of buffer use of, colorimeter / colour chart use of, balance / AW, to measure mass of precipitate method of maintaining temperature 9 repeat investigation twice / at least two replicates ; 10 relevant safety precaution – e.g. gloves / goggles / tongs / test-tube holder ;	6	

Question	Answer	Marks	Guidance
1(b)	OR disappearance of starch method <i>independent variable</i> 1 at least three temperature values, tested / stated ; <i>dependent variable</i> 2 time for iodine solution to stay yellow-brown / time for disappearance of starch / colour of iodine (solution) after a set period of time ; 3, 4 and 5 variables kept constant ;;; concentration of, amylase / enzyme (solution) volume / mass, of, amylase / enzyme (solution) pH volume of starch (solution) / mass of starch / mass of plant material / size of plant material concentration of starch (solution) 6, 7 and 8 method ;;; equilibration (both solutions in the water bath before mixing) take samples (testing with) iodine (solution) use of, spotting / white, tile use of buffer use of, colorimeter / colour chart method of maintaining temperature 9 repeat investigation twice / at least two replicates ; 10 relevant safety precaution – gloves / goggles ;		

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Question	Answer	Marks	Guidance
2(a)(i)	<i>any one from:</i> to get a representative sample ; to avoid bias ; to, identify / exclude, anomalies / outliers ; to increase confidence in results / to increase reliability / to check reliability ;	1	
2(a)(ii)	value indicated on graph ; 3.3 (m ²) ;	2	
2(a)(iii)	57(%) ;;;	3	MP1 for correct values selected from graph (1.4 and 2.2) or 0.8 MP2 calculation of percentage increase to any number of significant figures MP3 answer given to two significant figures ecf from previous step
2(b)	59 ; cm ² ;	2	A any value within the range of 55 to 66
2(c)	<i>method:</i> (add) Benedict's (reagent) ; heat ; <i>positive test:</i> green / yellow / orange / (brick) red / red-brown / brown ;	3	
2(d)	outline – single clear unbroken line ; size – at least 65mm wide ; detail 1: five lobes ; detail 2: one main leaf vein in each lobe ;	4	

Question	Answer	Marks	Guidance												
2(e)	<i>similarity:</i> lobed (AW) / number of lobes (AW) / presence of veins / pattern of veins / uneven edge ; <i>difference:</i> <table><tr><td>grapevine leaf / fig. 2.2 / first leaf</td><td></td><td>fig leaf / fig. 2.3 / second leaf</td></tr><tr><td>serrated / pointy / sharp / rough / ragged</td><td>vs</td><td>smooth / wavy / rounded</td></tr><tr><td>no stalk / no petiole</td><td>vs</td><td>stalk / petiole</td></tr><tr><td>fewer veins</td><td>or</td><td>more veins</td></tr></table> ;	grapevine leaf / fig. 2.2 / first leaf		fig leaf / fig. 2.3 / second leaf	serrated / pointy / sharp / rough / ragged	vs	smooth / wavy / rounded	no stalk / no petiole	vs	stalk / petiole	fewer veins	or	more veins	2	
grapevine leaf / fig. 2.2 / first leaf		fig leaf / fig. 2.3 / second leaf													
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